

Mathematics for Mechatronics Engineering

Linear Algebra: Intuition and Applications

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Session Series: 1 – 3

- **Session 1:** Vectors & Systems of Equations
- **Session 2:** Matrices & Transformations
- **Session 3:** Eigenvalues & Engineering Applications

Goal: Building engineering intuition over rigorous proofs.

Vectors: Engineering Intuition

Vectors are the language of physical space in Mechatronics.

Physical Interpretation

- **Position:** Coordinate in 2D/3D space.
- **Velocity:** Heading and magnitude of a mobile robot.
- **Force/Torque:** Directional effort applied to joints.

$$\vec{v} = \begin{bmatrix} x \\ y \\ z \end{bmatrix}$$

Solving Engineering Models ($Ax = b$)

Many mechatronic systems are modeled as systems of equations.

- **A:** The system properties (stiffness, resistance, geometry).
- **x:** The unknowns (joint forces, node voltages).
- **b:** The inputs (external loads, source currents).

Solving for x allows us to simulate and predict system behavior.

Matrices as Operators: Mapping Space

Instead of viewing a matrix as static data, mechatronics engineers view a matrix A as a **machine that transforms an input vector $\vec{x}$ into a new output vector $\vec{y}$** .

$$\vec{y} = A\vec{x}$$

- **Rotation:** Orthonormal columns map directions to new directions ($\det(A) = 1$).
- **Scaling:** Diagonal elements scale individual coordinate axes independently.
- **Translation:** Rigid-body displacement (requires non-linear mapping or higher dimensions).

Rotation Matrices $SO(2)$: Deep Dive

A rotation matrix in 2D space maps a vector from a moving frame $\{B\}$ back to a global reference frame $\{A\}$ by an angle θ :

$$R(\theta) = \begin{bmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{bmatrix}$$

Properties crucial for code verification:

- **Orthogonality:** $R^T R = I \implies R^{-1} = R^T$. (Undoing a rotation is just a matrix transpose—extremely efficient computationally!).
- **Preservation:** Lengths of vectors and angles between them never change under R .

Example 1: Rotating a Robotic Link Vector

Problem: A planar robotic arm link has a point at position $\vec{p}_B = [2, 1]^T$ relative to its joint axis. The joint rotates by $\theta = 30^\circ$. Find its coordinates $\vec{p}_A$ in the base frame.

Calculation:

$$R(30^\circ) = \begin{bmatrix} \cos(30^\circ) & -\sin(30^\circ) \\ \sin(30^\circ) & \cos(30^\circ) \end{bmatrix} = \begin{bmatrix} 0.866 & -0.5 \\ 0.5 & 0.866 \end{bmatrix}$$

$$\vec{p}_A = R(30^\circ)\vec{p}_B = \begin{bmatrix} 0.866 & -0.5 \\ 0.5 & 0.866 \end{bmatrix} \begin{bmatrix} 2 \\ 1 \end{bmatrix}$$

$$\vec{p}_A = \begin{bmatrix} (0.866 \cdot 2) + (-0.5 \cdot 1) \\ (0.5 \cdot 2) + (0.866 \cdot 1) \end{bmatrix} = \begin{bmatrix} 1.232 \\ 1.866 \end{bmatrix}$$

Homogeneous Transformations $SE(3)$

Pure matrix multiplication cannot perform vector translation ($\vec{x} + \vec{t}$) linearly. To combine **Rotation** (R) and **Translation** ($\vec{t}$) into a single multiplication operator, we augment our vectors into 4D space.

The 4×4 Homogeneous Transformation Matrix (T):

$$T = \begin{bmatrix} R_{3 \times 3} & \vec{t}_{3 \times 1} \\ 0_{1 \times 3} & 1 \end{bmatrix} = \begin{bmatrix} r_{11} & r_{12} & r_{13} & x \\ r_{21} & r_{22} & r_{23} & y \\ r_{31} & r_{32} & r_{33} & z \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Vectors are appended with a scaling factor $w = 1$: $\vec{p} = [x, y, z, 1]^T$.

Example 2: 2D Homogeneous Setup

Problem: A camera frame $\{C\}$ is rotated by 90° and translated by $x = 5, y = 2$ relative to a global frame $\{G\}$. Set up its matrix T and locate an object found at local position $\vec{p}_C = [0, 1, 1]^T$.

Calculation:

$$R(90^\circ) = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}, \quad \vec{t} = \begin{bmatrix} 5 \\ 2 \end{bmatrix} \implies {}^G T_C = \begin{bmatrix} 0 & -1 & 5 \\ 1 & 0 & 2 \\ 0 & 0 & 1 \end{bmatrix}$$

$$\vec{p}_G = {}^G T_C \cdot \vec{p}_C = \begin{bmatrix} 0 & -1 & 5 \\ 1 & 0 & 2 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}$$

$$\vec{p}_G = \begin{bmatrix} (0 \cdot 0) + (-1 \cdot 1) + (5 \cdot 1) \\ (1 \cdot 0) + (0 \cdot 1) + (2 \cdot 1) \\ 1 \end{bmatrix} = \begin{bmatrix} 4 \\ 2 \\ 1 \end{bmatrix} \implies (x_G = 4, y_G = 2)$$

Forward Kinematics: Chaining Operators

For serial multi-joint industrial robots, every link frame depends directly on the physical state of the preceding joint frame.

- Let ${}^0T_1(\theta_1)$ be the transformation from the base to joint 1.
- Let ${}^1T_2(\theta_2)$ be the transformation from joint 1 to joint 2.

To find a tool point's absolute location relative to the floor frame (0), we chain by multiplying matrices sequentially from left to right:

$${}^0T_{tool} = {}^0T_1(\theta_1) \cdot {}^1T_2(\theta_2) \cdot \dots \cdot {}^{n-1}T_n(\theta_n)$$

CRITICAL Order Alert: Matrix multiplication is non-commutative ($A \cdot B \neq B \cdot A$). Reversing the sequence breaks your robot arm simulation completely!

Example 3: 2-Link Robotic Arm Kinematics

Problem: A flat 2-link manipulator arm has link lengths $L_1 = 10$, $L_2 = 5$. If joint 1 is rotated at $\theta_1 = 0^\circ$ and joint 2 is at $\theta_2 = 90^\circ$, find the global position of the tool-tip $\vec{p}_{tip} = [5, 0, 1]^T$ relative to joint 2.

Step 1: Set up individual joint matrices

$${}^0T_1 = \begin{bmatrix} \cos(0^\circ) & -\sin(0^\circ) & 0 \\ \sin(0^\circ) & \cos(0^\circ) & 0 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$${}^1T_2 = \begin{bmatrix} \cos(90^\circ) & -\sin(90^\circ) & L_1 \\ \sin(90^\circ) & \cos(90^\circ) & 0 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 0 & -1 & 10 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Example 3: Compounding Matrices

Step 2: Chain matrices to build overall Forward Kinematics

$${}^0T_2 = {}^0T_1 \cdot {}^1T_2 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 0 & -1 & 10 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 0 & -1 & 10 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Step 3: Solve for End-Effector Absolute Position

$$\vec{p}_{base} = {}^0T_2 \cdot \vec{p}_{tip} = \begin{bmatrix} 0 & -1 & 10 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 5 \\ 0 \\ 1 \end{bmatrix}$$

$$\vec{p}_{base} = \begin{bmatrix} (0 \cdot 5) + (-1 \cdot 0) + (10 \cdot 1) \\ (1 \cdot 5) + (0 \cdot 0) + (0 \cdot 1) \\ 1 \end{bmatrix} = \begin{bmatrix} 10 \\ 5 \\ 1 \end{bmatrix}$$

The robotic hand is at coordinates ($x = 10, y = 5$) in the room space.

Eigenvalues: The System's DNA

An eigenvector is a direction that doesn't change during a transformation.
An eigenvalue λ is the scaling factor along that direction.

$$A\vec{v} = \lambda\vec{v}$$

In mechatronics, these values tell us the "natural" behavior of a system.

Consider a state-space model: $\dot{x} = Ax + Bu$. The stability of the controller depends on the eigenvalues of matrix A .

- **Negative Real Parts:** The system stabilizes.
- **Positive Real Parts:** The system explodes/oscillates (unstable).

How to Find Eigenvalues: The Algorithm

To find the eigenvalues λ of a square matrix A , we follow these three logical steps:

- 1 **Form the Characteristic Matrix:** Subtract λ from the diagonal of A to get $(A - \lambda I)$.
- 2 **Calculate the Determinant:** Find $|A - \lambda I|$ to create a polynomial in terms of λ .
- 3 **Solve for Roots:** Set the polynomial to zero ($\det(A - \lambda I) = 0$) and solve for λ .

The Characteristic Equation

The equation $\det(A - \lambda I) = 0$ is the "DNA" of your system.

- For a 2×2 matrix, it results in a quadratic equation:

$$\lambda^2 - \text{tr}(A)\lambda + \det(A) = 0$$

- The roots of this equation are the eigenvalues.
- In mechatronics, these roots tell us if the robot arm oscillates, drifts, or stays put.

Example: Mass-Spring-Damper System

Consider a simple mechanical system with $m = 1$, $k = 2$, $c = 3$. The state-space matrix A (relating position and velocity) is:

$$A = \begin{bmatrix} 0 & 1 \\ -2 & -3 \end{bmatrix}$$

Step 1: Form $(A - \lambda I)$

$$A - \lambda I = \begin{bmatrix} -\lambda & 1 \\ -2 & -3 - \lambda \end{bmatrix}$$

Example: Step-by-Step Calculation

Step 2: Calculate the Determinant

$$\begin{aligned}\det(A - \lambda I) &= (-\lambda)(-3 - \lambda) - (1)(-2) \\ &= 3\lambda + \lambda^2 + 2\end{aligned}$$

Step 3: Solve for λ

$$\lambda^2 + 3\lambda + 2 = 0 \implies (\lambda + 1)(\lambda + 2) = 0$$

The eigenvalues are:

$$\lambda_1 = -1, \quad \lambda_2 = -2$$

What do $\lambda_1 = -1$ and $\lambda_2 = -2$ mean for our robot?

PHYSICAL INTERPRETATION

- **Stability:** Both values are **negative**. This means the system is stable and will always return to its resting position.
- **Damping:** Because the values are **real** (no imaginary i parts), the system does not oscillate. It is "overdamped."
- **Response Time:** The system decays at rates e^{-t} and e^{-2t} . The -1 eigenvalue is the "slowest" pole, dominating the response.

Summary of Applications

- **Robotics:** FK/IK and Jacobian analysis.
- **Signal Processing:** Principle Component Analysis (PCA).
- **Vibrations:** Modal analysis of mechanical structures.